Observation Field Sheet

650 nm laser diffraction observation protocol | Structured recording for the DMT Code research registry

BEFORE YOU BEGIN

This field sheet standardizes how a laser diffraction observation session is recorded so that independent sessions can be compared. Complete one sheet per session. Fields left blank reduce the comparability of your record. Submit completed records at dmtcode.com/registry. Adults 18+ only. Read the safety card on page 4 before setup.

The observation was first described by Danny Goler in August 2020 and written up as a pilot study in IPI Letters (DOI 10.59973/ipil.158); he has no part in Meridian Optics Lab, the store or the site, is not a founder, holds no editorial role and endorses nothing sold or published here.

STANDARD APPARATUS

1. 650 nm red laser pointer (Class IIIa / 3R, at or below 5 mW, the Class 3R ceiling; the kit pointers are vendor rated 5 mW, FDA Class IIIa), mounted on an adjustable stand. Hands-free mounting is required: handheld beams move with pulse and breath and contaminate the record.
2. Diffraction grating positioned in the beam path, close to the aperture. The grating fans the single beam into an extended spread of diffraction orders on the projection surface. This spread is the observation field. The $199 Protocol kit ships a glass rod in place of a grating, a cylindrical optic that refracts the dot into a single line and produces no diffraction orders; in that case record the Grating in beam path field as none, glass rod, and leave the diffraction line fields as none.
3. Projection surface: a flat, light-colored, matte wall or screen, 3 to 5 meters from the laser (site protocol, wider than the paper; the pilot study places the surface 4 to 6 ft, 1.2 to 1.8 m, from the laser).
4. Room darkened. The pattern should be the brightest object in the visual field.

STANDARD ALIGNMENT PROCEDURE

1. Place the stand on a stable surface. Aim the beam at the projection surface, roughly perpendicular.
2. Insert the grating into the beam path. Confirm the projected pattern spreads into an extended line or array.
3. Adjust distance until the spread pattern is 0.5 to 1.5 meters wide on the surface (site protocol, wider than the paper).
4. Confirm no reflective objects (mirrors, glass, metal, screens) intersect the beam path.
5. Seat observers 1 to 3 meters from the surface (site protocol, wider than the paper), viewing the projected pattern only. Never view the aperture.
6. Record the setup fields on page 2 before the observation period begins.

RECORDING DISCIPLINE

Record only what is directly observed in the projected pattern. Distinguish between description (what appeared) and interpretation (what you think it means); this sheet records description only. Sketch immediately, from observation where possible, from short-term recall otherwise, and note which. If multiple observers are present, each observer completes their own sheet without conferring until all sheets are complete. Independent records that agree are evidence. Records harmonized by discussion are not.

Session Record

Complete before and during the observation period. One sheet per observer per session.

SESSION

Date _____ Start time / end time _____

Location type (home, studio, outdoor, other) _____ Observer ID or initials (optional) _____

Observers present (count) _____ Session number in your series _____

APPARATUS CONFIGURATION

Laser model / source _____ Wavelength (nm) _____

Grating in beam path (type / lines per mm; or glass rod) _____ Laser-to-surface distance (m) _____

Diffraction lines (count, spacing, brightness; or none) _____ Optic-to-laser distance (cm) _____

Projected pattern width on surface (m) _____ Observer-to-surface distance (m) _____

Projection surface (wall paint, screen, other) _____ Ambient light (fully dark, dim, other) _____

Corrective lenses worn (none, glasses, contacts) _____ Mount stable and hands-free (Y/N) _____

PATTERN CHARACTER, CHECK ALL OBSERVED

- ☐ Static geometric structure
- ☐ Repeating lattice or grid
- ☐ Discrete glyph-like elements
- ☐ Apparent motion or flow
- ☐ Depth or layering
- ☐ Color fringing beyond red
- ☐ Elements stable on re-inspection
- ☐ Elements changed on re-inspection
- ☐ No structure beyond speckle

TIMING

Time from session start to first structured observation _____ Duration structure remained observable _____

OBSERVATION NOTES, DESCRIPTION ONLY

Sketch Record

Sketch observed elements immediately. Note whether drawn during observation or from recall.

Element 1

During observation [ ] From recall [ ]

Element 2

During observation [ ] From recall [ ]

Element 3

During observation [ ] From recall [ ]

Element 4

During observation [ ] From recall [ ]

Element 5

During observation [ ] From recall [ ]

Element 6

During observation [ ] From recall [ ]

FULL-FIELD SKETCH, THE PROJECTED PATTERN AS A WHOLE

Confidence that sketches reflect what was observed (low / moderate / high) and why _____

Laser Safety Card

Keep with the apparatus. Review before every session.

THE INSTRUMENT

This kit uses a Class IIIa / 3R visible laser, at or below 5 mW output at 650 nm, the Class 3R ceiling, compliant with US FDA CDRH requirements for consumer laser products. The kit pointers are vendor rated 5 mW, FDA Class IIIa (Class 3R). The pilot study's own instrument was a 650 nm Class 2 laser at 1 mW, quoted here for comparison only; the kits are not that setup. At this class, the blink reflex normally protects against momentary accidental exposure, but deliberate or prolonged direct viewing can cause permanent eye injury.

ABSOLUTE RULES

- Never look into the laser aperture, powered or not.
- Never point the beam at eyes, faces, people, or animals.
- Never view the beam or its reflections through magnifying optics, binoculars, cameras with optical viewfinders, or any lens system. Optics concentrate the beam and defeat the eye's natural protection.
- Never direct the beam at vehicles or aircraft. Doing so is a federal crime in the United States.
- Never leave the laser powered and unattended.
- Remove reflective objects (mirrors, glass, polished metal, dark screens) from the beam path before powering on.
- Adults 18+ operate the laser. Keep it out of the reach of children.

SESSION HYGIENE

Observers view the projected pattern on the surface, never the aperture and never specular reflections. Power the laser off between observation periods. Confirm stable mounting before each power-on so the beam cannot swing across the room if the stand is bumped. If any observer reports afterimages persisting more than a few minutes, end the session; persistent visual changes after any suspected direct exposure warrant a prompt eye examination.

CARE

Store the laser with batteries removed. Keep the grating clean and dry; handle it by the frame edges only, since surface oils degrade the diffraction pattern. Do not disassemble the laser housing.

Acknowledgment

I have read this safety card and will follow it during every session.

Signature _____ Date _____